Suppose $X \sim N(0, \sigma^2)$. You want to use Jeffrey's prior for σ. Thus,

$$L(\sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{x^2}{\sigma^2}}.$$

$$l(\sigma) = c - \log(\sigma) - \frac{1}{2} \cdot \frac{x^2}{\sigma^2}.$$

$$\frac{\partial l(\sigma)}{\partial \sigma} = -\frac{1}{\sigma} \cdot - \frac{1}{2} \cdot x^2 \cdot (-2 \cdot \sigma^{-3})$$

$$= -\frac{1}{\sigma} + \frac{1}{\sigma^3} x^2.$$

$$\frac{\partial^2 l(\sigma)}{\partial \sigma^2} = \frac{1}{\sigma^2} - 3x^2 \cdot \frac{1}{\sigma^4}.$$

$$E\left(-\frac{\partial^2 l(\sigma)}{\partial \sigma^2}\right) = E\left(-\frac{1}{\sigma^2} + 3 \cdot \frac{x^2}{\sigma^4}\right)$$

$$= -\frac{1}{\sigma^2} + 3 \cdot \frac{1}{\sigma^4} \cdot E(x^2)$$

$$= -\frac{1}{\sigma^2} + 3 \cdot \frac{\sigma^2}{\sigma^4} = \frac{2}{\sigma^2}.$$

$$I(\sigma) = \frac{2}{\sigma^2} \Rightarrow \pi(\sigma) \propto \sqrt{I(\sigma)} \propto \sigma^{-1}$$

Thus, you use the prior $\pi_1(\sigma) \propto \sigma^{-1}$.

I want to use Jeffrey's prior for $\phi = \sigma^2$. Thus,

$$L(\phi) = \frac{1}{\sqrt{2\pi\phi}} e^{-\frac{1}{2} \cdot \frac{x^2}{\phi}}$$

$$l(\phi) = -\frac{1}{2}\log(\phi) - \frac{1}{2}\frac{x^2}{\phi}.$$

$$l'(\phi) = -\frac{1}{2} \cdot \frac{1}{\phi} + \frac{1}{2}\frac{x^2}{\phi^2}$$

$$\ell''(\phi) = + \frac{1}{2} \cdot \frac{1}{\phi^2} - \frac{1}{\cancel{2}} \cdot \frac{\cancel{2} x^2}{\phi^3} = \frac{1}{2\phi^2} - \frac{x^2}{\phi^3}$$

$$E(-\ell''(\phi)) = -\frac{1}{2\phi^2} + \frac{1}{\phi^3} \cdot E(x^2) = -\frac{1}{2\phi^2} + \frac{\phi}{\phi^3}$$

$$= \frac{1}{2\phi^2}$$

$$I(\phi) = \frac{1}{2\phi^2} \Rightarrow \pi(\phi) \propto \sqrt{I(\phi)} \propto \phi^{-1}.$$

Thus, $\pi_2(\phi) \propto \phi^{-1}$.

Now, π_3 is the prior obtained for ϕ, from your choice of π_1 for σ.

Thus, $\pi_3(\phi)\, d\phi = \pi_1(\sigma)\, d\sigma$. $\qquad \sigma = \sqrt{\phi}$

$$\Rightarrow \pi_3(\phi) = \pi_1(\sigma) \times \frac{d\sigma}{d\phi}$$

$$\Rightarrow \pi_3(\phi) = \pi_1(\sigma) \times \frac{d}{d\phi}\sqrt{\phi}$$

$$\Rightarrow \pi_3(\phi) \propto \frac{1}{\sigma} \cdot \frac{1}{2}\phi^{-1/2} \qquad \sigma = \sqrt{\phi}$$

$$\Rightarrow \pi_3(\phi) \propto \frac{1}{\sqrt{\phi}} \cdot \phi^{-1/2}$$

$$\Rightarrow \pi_3(\phi) \propto \phi^{-1}$$

Thus, $\pi_2 = \pi_3$. and thus, Jeffrey's prior is invariant to parameterization.